NLP Risk Engine

From unstructured news & social text to actionable financial risk signals

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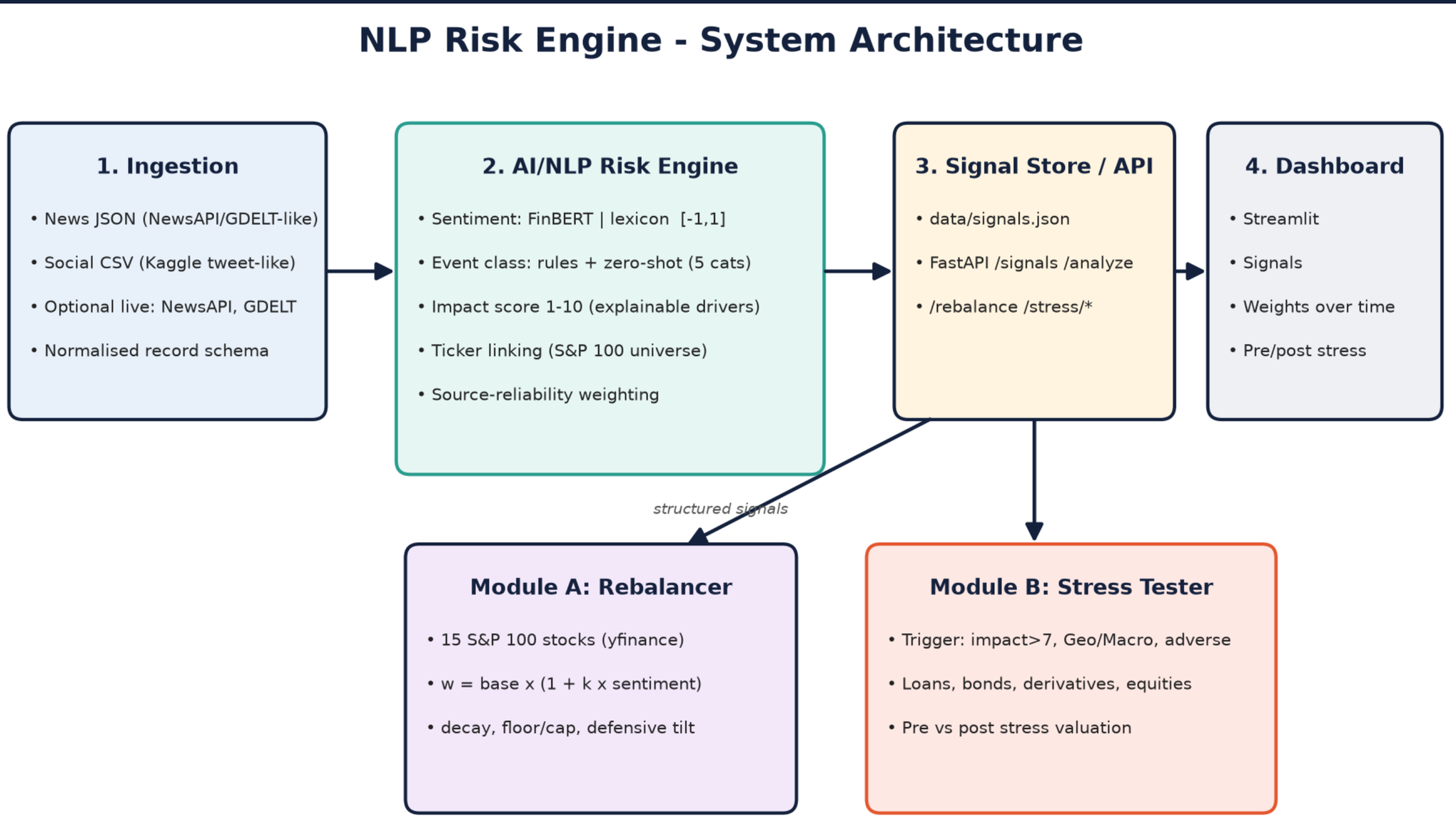
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S&P Global & Crisil Campus Hackathon 2026 | Modules A + B

Problem & Approach

- **Problem:** Risk-moving information arrives as unstructured text (headlines, tweets). Humans cannot read, score and act on it at market speed.
- **Approach:** one NLP engine turns every text into a machine-readable signal: sentiment [-1,1], event category (5 types) and impact score [1-10], linked to S&P 100 tickers.
- **Two consumers of the same signals:** Module A (tactical rebalancing of 15 stocks) and Module B (strategic stress test of a synthetic wholesale-banking book).
- **Design principles:** explainable, runs offline in seconds, swaps in FinBERT for production-grade accuracy.

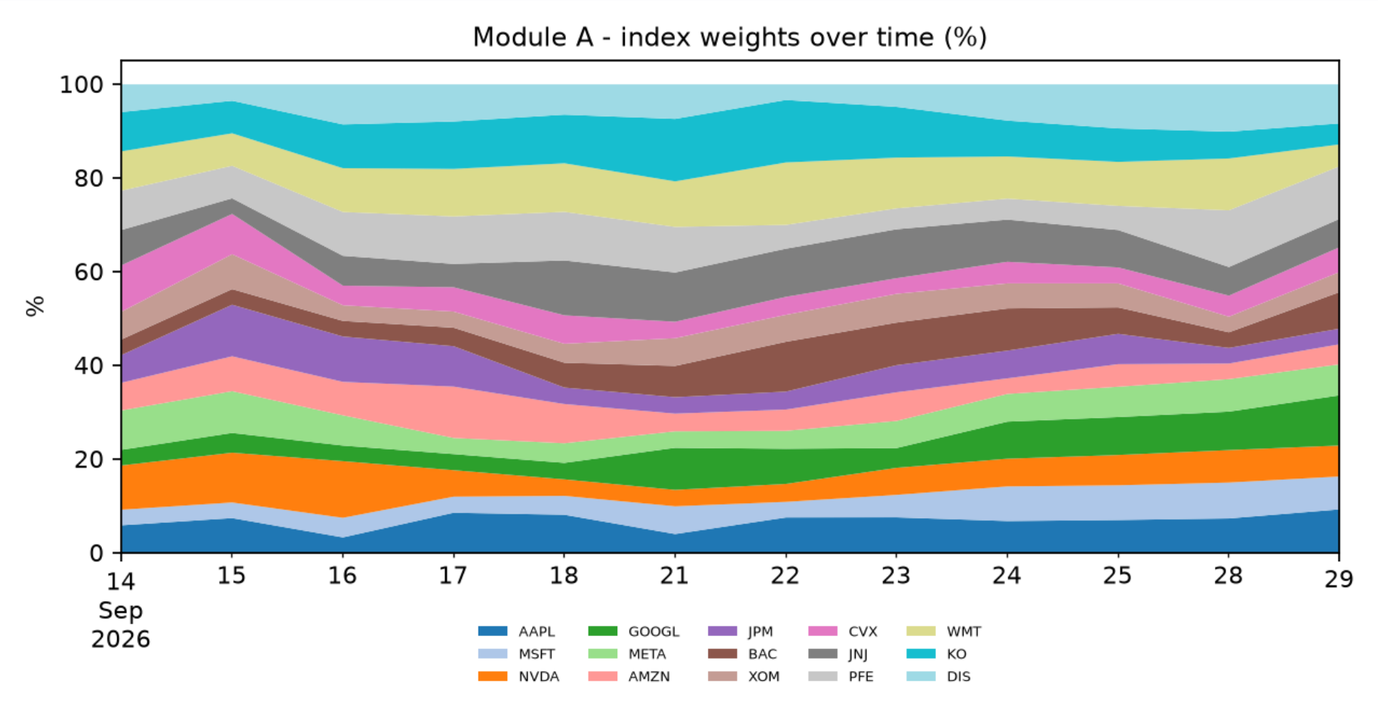
System Design

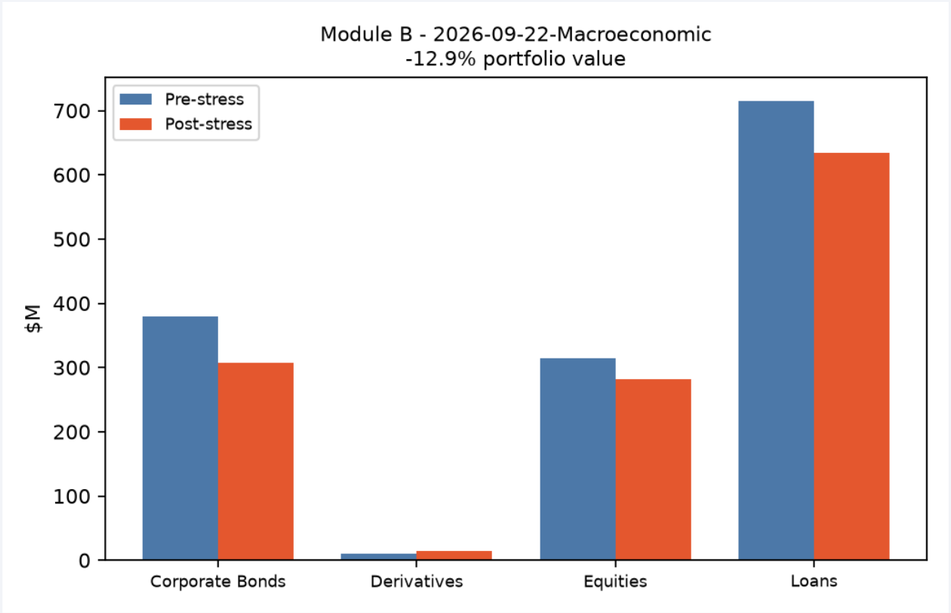


Implementation Highlights

Module	Tech choice	Why
Ingestion	JSON + CSV loaders; NewsAPI / GDELT live fetchers	2 sources, one schema; works offline
Sentiment	Finance lexicon (default) | FinBERT (opt-in)	Instant & dependency-free; FinBERT for accuracy
Event class	Weighted rules + BART zero-shot fallback	Explainable, 5 required categories
Impact 1-10	Additive model: category, severity, |sentiment|, scope	Every score ships with its drivers
Module A	w = base x (1 + k x s), decay, floor/cap	Bounded, smooth, no look-ahead
Module B	Duration/convexity, PD/LGD, DV01, delta, CS01	Simple, auditable shock model
Serving	FastAPI (async) + Streamlit + pytest (13 tests)	REST signals + interactive dashboard

Key Results





- **152 texts** (news + social) scored in **0.10s** on CPU; **10** stress events triggered; worst case **-12.9%** of book value.
- Held-out hand-written set (n=30): baseline **57%** sentiment / **77%** category; after lexicon expansion **87%** / **97%** (set then used for tuning = dev set). Template data scores 100% (not meaningful).
- Demo prices are synthetic, so index-vs-equal-weight is illustrative only; weight mechanics verified by unit tests.

Domain Impact

- **Asset managers:** turns news flow into bounded, auditable weight tilts (floor/cap, turnover metrics) instead of ad-hoc discretionary reactions.
- **Risk officers / banks:** geopolitical or macro shocks automatically trigger a stress run of the book, with pre/post values by asset class, in seconds rather than a next-day batch.
- **Efficiency vs manual tracking:** corroborating news + tweets are clustered into a single event (no alert fatigue); every score carries explainable drivers for audit.
- **Extensible:** same signal API can feed limits monitoring, credit watchlists or surveillance.

Limitations & Next Steps

- **Synthetic data:** bundled news, tweets, prices and portfolio are synthetic; no real or client data used.
- **Default NLP is a lexicon**; FinBERT/zero-shot are wired in but opt-in. Impact score is heuristic, not calibrated against realised market moves.
- **Stress model is simplified** (parallel shifts, no correlations, no counterparty/liquidity effects).
- **Next:** fine-tune FinBERT on labelled finance text, calibrate impact on event-study returns, streaming ingestion (Kafka), intraday rebalancing with transaction costs, multi-asset & reverse stress tests.